## Apache Airflow Case Study Documentation

### 1. Introduction

Apache Airflow is an open-source platform to programmatically author, schedule, and monitor workflows. It allows building complex data pipelines using Directed Acyclic Graphs (DAGs) and provides a web interface to manage and monitor workflow execution.

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### 2. Key Features of Airflow

- **DAG-based Workflows**: Pipelines are defined as Directed Acyclic Graphs for better visualization and management.
- **Dynamic Pipeline Generation**: Workflows are written in Python, enabling dynamic creation of tasks.
- **Task Scheduling**: Provides robust scheduling capabilities with cron-like expressions.
- **Scalability**: Can scale horizontally across multiple workers for handling large workloads.
- **Extensible**: Supports custom operators, sensors, and hooks to integrate with external systems.
- **Monitoring and Logging**: Built-in web interface for monitoring DAG runs, task logs, and execution status.
- **Retry and Alerting**: Configurable retries, alerts, and error handling for reliable execution.
- **Integration**: Works with multiple data sources, cloud services, and orchestration tools.

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### 3. Building a Pipeline in Airflow

To build and run a pipeline in Airflow, follow these steps:

#### Step 1: Define a DAG File

1. Open the Airflow project directory.
2. Locate the dags folder.
3. Create a new Python file (e.g., example_dag.py).

4. Import the required Airflow libraries such as DAG, operators, and datetime.
5. Define the default arguments like owner, start date, retries, etc.
6. Initialize a DAG object with a unique DAG ID and scheduling interval.
7. Define tasks using available operators (e.g., BashOperator, PythonOperator).
8. Set dependencies between tasks using >> or <<.

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### Step 2: Place the DAG in the Correct Folder

- Save the Python file in the dags directory.
- Ensure the file has no syntax errors, as Airflow scans this folder to detect DAGs.

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### Step 3: Start Airflow Services

1. Start the Airflow web server.
2. Start the Airflow scheduler.
3. Ensure both services are running in the background.

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### Step 4: Access Airflow UI

1. Open a browser and go to the Airflow web interface (default: http://localhost:8080).
2. Log in using the default or configured credentials.
3. Navigate to the **DAGs page**, where all available pipelines will be listed.

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### Step 5: Enable and Run a DAG

1. Locate the newly created DAG in the list.
2. Toggle the switch to enable the DAG.
3. Trigger the DAG manually or wait for the scheduled run.

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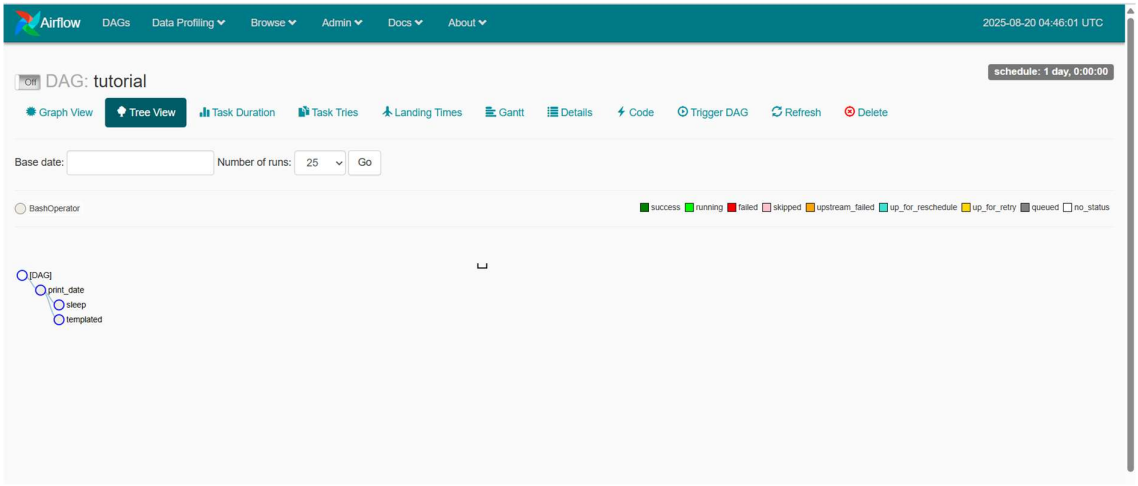
### Step 6: Monitor Pipeline Execution

1. Click on the DAG name to view details.
2. Use the **Graph View** to visualize task dependencies.
3. Use the **Tree View** to track execution history.
4. Open **Task Logs** to view real-time logs of each task execution.

### Sample Screenshot of Tree View:

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### 4. Viewing and Managing Pipelines

- **DAGs List View**: Shows all available workflows with their current status.
- **Graph View**: Visualizes the structure of the DAG and task dependencies.
- **Tree View**: Displays task run history over multiple DAG runs.
- **Task Instance Details**: Provides logs, retry options, and execution information.
- **Code View**: Displays the underlying DAG Python code directly in the UI.

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### 5. Best Practices

- Use clear and descriptive DAG and task names.
- Keep DAG files lightweight and modular.
- Define retries and failure handling mechanisms.
- Use variables and connections for configurable settings.

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